

ZHIQI HUANG

SHENZHEN, CHINA / FUKUOKA, JAPAN

EMAIL HOMEPAGE SCHOLAR GITHUB

RESEARCH PROFILE

- Research Assistant at CUHK-Shenzhen and M.Phil. student at Waseda University, working on world models for embodied AI with a focus on understanding and reducing the sim-to-real gap.
- My background spans production graphics, controllable 3D generation, physically based appearance, robust perception, and structured prediction—foundations I now bring to world-model-based robot learning.

RESEARCH INTERESTS

- World models for embodied AI and robotic manipulation.
- Sim-to-real transfer through reality-gap diagnosis and adaptation.
- Learning from simulated and real experience for robust policy training and evaluation.

RESEARCH EXPERIENCE

The Chinese University of Hong Kong, Shenzhen

Jul. 2026 - Present

Research Assistant

Shenzhen, China

- Research focus: using world models to support embodied AI, with particular emphasis on understanding and reducing the sim-to-real gap.

EDUCATION

Waseda University

Apr. 2025 - Mar. 2027 (expected)

M.Phil. in Information Architecture

Fukuoka, Japan

- English-taught program; GPA: 3.8 / 4.0

Sun Yat-sen University

2017 - 2021

B.E. in Software Engineering

Guangzhou, China

- GPA: 3.7 / 4.0

PUBLICATION

SIE3D: Single-Image Expressive 3D Avatar Generation via Semantic Embedding and Perceptual Expression Loss 2026

IEEE ICASSP 2026; First Author, Corresponding Author

- Authors: Zhiqi Huang, Dulongkai Cui, Jinglu Hu
- Generates an editable 3D Gaussian head avatar from one image while preserving identity and enabling language-level control over expressions and accessories.
- Links: Project, IEEE Xplore, arXiv, Code

MANUSCRIPTS UNDER REVIEW**Controllable PBR material generation from long-form descriptions 2026**

First Author; AAAI 2027 (Under Review)

- Generates relightable PBR channels by retaining detailed material specifications and routing them with surface geometry.
- Extends controllable 3D modeling to geometry-aligned, physically based object appearance, with relevance to the visual-observation side of real-to-sim.

Deform-then-edit forecasting for longitudinal 3D CT 2026

First Author; AAAI 2027 (Under Review)

- Forecasts localized volumetric change while explicitly preserving stable surrounding anatomy.
- Introduces a history-conditioned deform-then-edit transition model; it provides a methodological basis for future action-conditioned 3D transitions but does not model robot actions itself.

Robust pseudo-labeling under imaging noise and long-tailed data 2026

Second Author, Corresponding Author; NeurIPS 2026 (Under Review)

- Develops feature-threshold dual calibration for rare-class semantic segmentation across sonar, underwater, and adverse-weather imagery.
- Contributes robust semantic perception for simulation-ready world representations under sensor and distribution shifts.

INDUSTRY EXPERIENCE**4399 Games 2023 - 2024**

Senior Graphics Engineer

Guangzhou, China

- Led a 3-5 person rendering team and owned the rendering roadmap for *Era of Conquest* on mobile and PC.
- Developed reusable asset, material, and rendering pipelines for mobile, PC, and web projects.

4399 Games 2021 - 2023

Graphics Engineer

Guangzhou, China

- Built and optimized the cross-platform real-time rendering pipeline for *Era of Conquest*, including shaders, PBR materials, and mobile performance profiling.

TECHNICAL SKILLS

- Graphics and systems: C++, C#, GLSL/HLSL; Unity, Vulkan, OpenGL, real-time rendering, PBR, cross-platform optimization.
- Machine learning and 3D: Python, PyTorch, 3D Gaussian Splatting, diffusion models, CLIP/LongCLIP, mesh and material processing, multi-view evaluation, volumetric modeling.

LANGUAGES AND HONORS

- Chinese: Native (Mandarin and Cantonese)
- English: Professional working proficiency (TOEFL iBT: 90)
- Outstanding Student Scholarship (Third Prize), Sun Yat-sen University, 2018-2019